Python-Pandas

Objective

- Basics of Pandas
- Python Pandas Data Structure
- Pandas Operation
- Pandas DataFrame
- Data Cleaning
- Summary

Basics of Pandas

- Pandas is defined as an open-source library that provides high-performance data manipulation in Python.
- The name of Pandas is derived from the word Panel Data, which means an Econometrics from Multidimensional data.
- It is used for data analysis in Python and developed by Wes McKinney in 2008.

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Benefits of Pandas

- **Data Representation:**

  It represents the data in a form that is suited for data analysis through its DataFrame and Series.

- **Clear code:**

  The clear API of the Pandas allows you to focus on the core part of the code. So, it provides clear and concise code for the user

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Installation

- pip install pandas

If you are using Anaconda

- Start Navigator.
- In Environments tab checked for installed packeges.

Python Pandas Data Structure

OBJECT TECHNOLOGIES

1. **Series :**

- It is defined as a one-dimensional array that is capable of storing various data types.
- The row labels of series are called the index.
- We can easily convert the list, tuple, and dictionary into series using "series' method.
- A Series cannot contain multiple columns
- Example

```
import pandas as pd
import numpy as np
info = np.array(['P','a','n','d','a','s'])
a = pd.Series(info)
print(a)
```


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DataFrame

OBJECT™ TECHNOLOGIES

2. DataFrame

- It is a widely used data structure of pandas and works with a two-dimensional array with labeled axes (rows and columns).
- DataFrame is defined as a standard way to store data and has two different indexes, i.e., row index and column index.
- The columns can be heterogeneous types like int, bool, and so on.
- Pandas DataFrame consists of three principal components, the **data**, **rows**, and **columns**.
- Example:

```
import pandas as pd
# a list of strings
x = ['Python', 'Pandas']

# Calling DataFrame constructor on list
df = pd.DataFrame(x)
print(df)
```


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Series

	apples
0	3
1	2
2	0
3	1

+

Series

	oranges
0	0
1	3
2	7
3	2

=

DataFrame

	apples	oranges
0	3	0
1	2	3
2	0	7
3	1	2

Pandas Operation on Series

OBJECT™ TECHNOLOGIES

data

- list
- Dictionary
- scalar value

index

- should be unique
- default **np.arrange(n)**

Dtype

- It refers to the data type of series

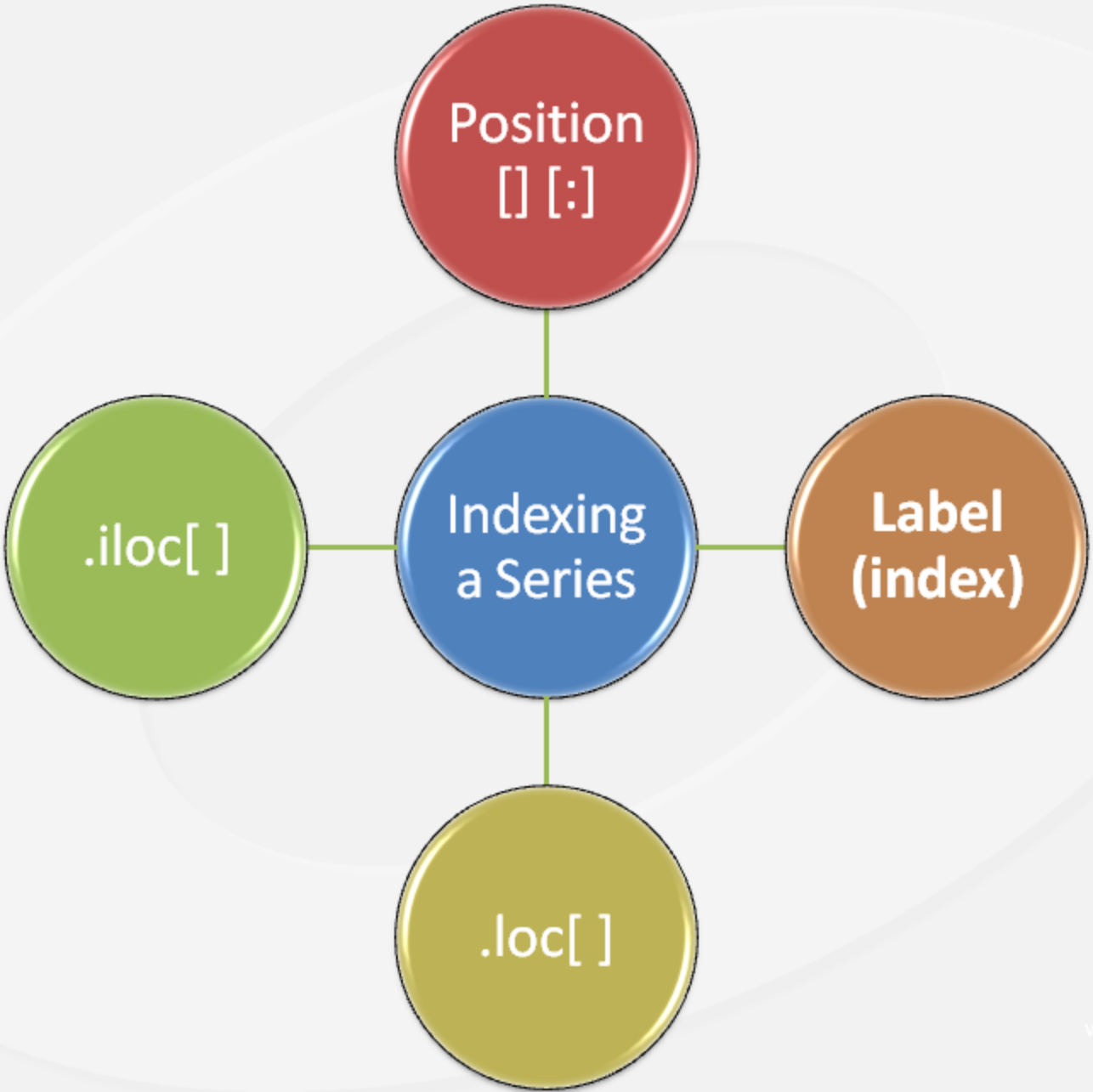
copy

- It is used for copying the data.

Creating Series

- using inputs
  - •Array
  - •Dict
  - •Scalar value
- Empty Series
  - 1.<series object> = pandas.Series()

Indexing a Series



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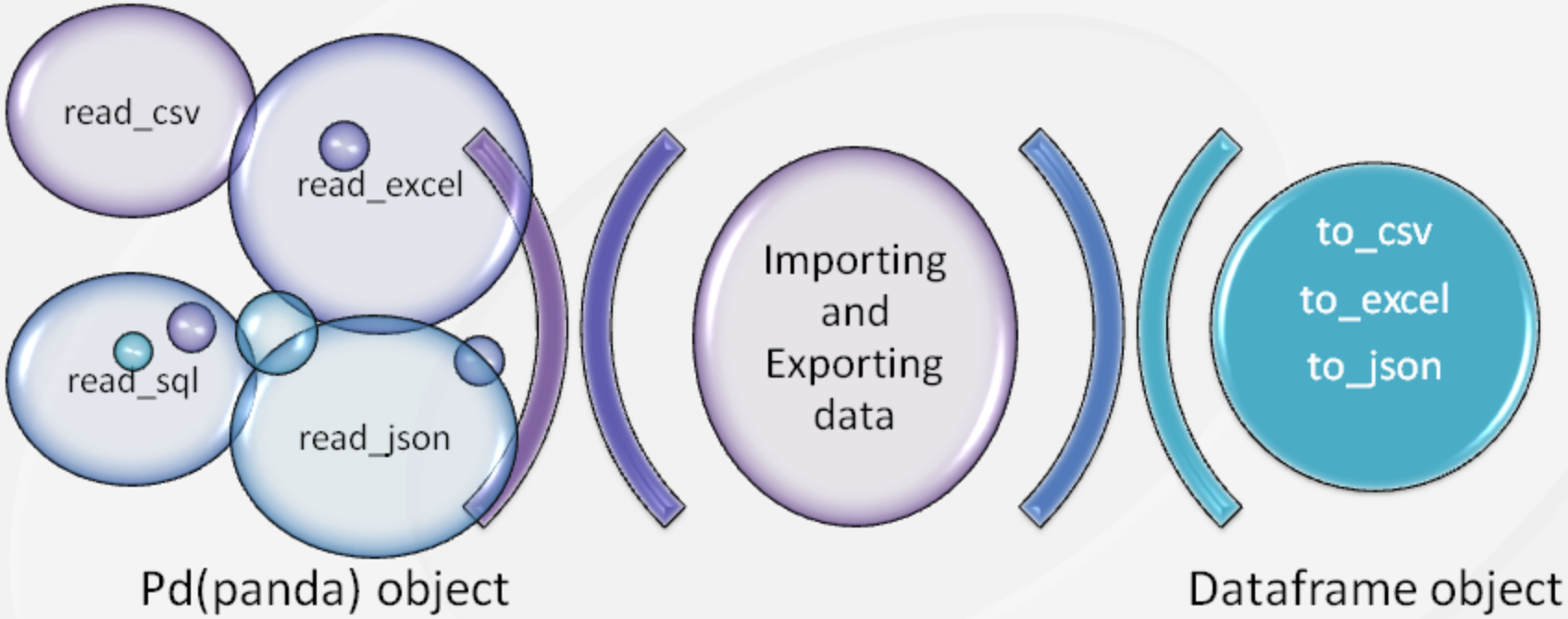
Operation on Series

OBJECT™ TECHNOLOGIES

FUNCTION	DESCRIPTION
add()	Method is used to add series or list like objects with same length to the caller series
sub()	Method is used to subtract series or list like objects with same length from the caller series
mul()	Method is used to multiply series or list like objects with same length with the caller series
div()	Method is used to divide series or list like objects with same length by the caller series
sum()	Returns the sum of the values for the requested axis
prod()	Returns the product of the values for the requested axis
mean()	Returns the mean of the values for the requested axis
pow()	Method is used to put each element of passed series as exponential power of caller series and returned the results
abs()	Method is used to get the absolute numeric value of each element in Series/DataFrame
cov()	Method is used to find covariance of two series
round()	Round is used to round the value
astype()	Method is used to change data type of a series
value_counts()	Method to count the number of the times each unique value occurs in a Series

Pandas DataFrame



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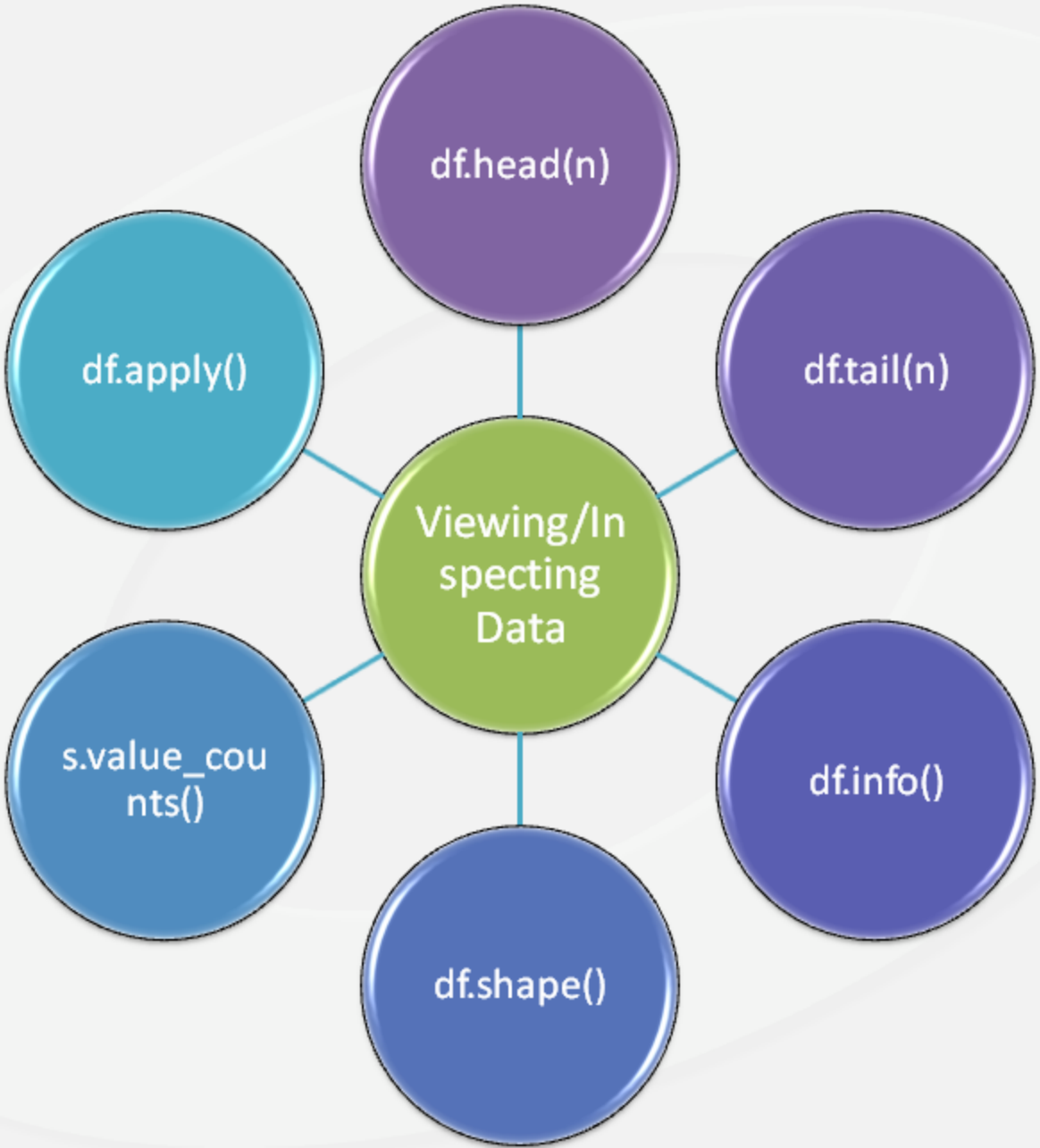
Viewing/Inspecting Data



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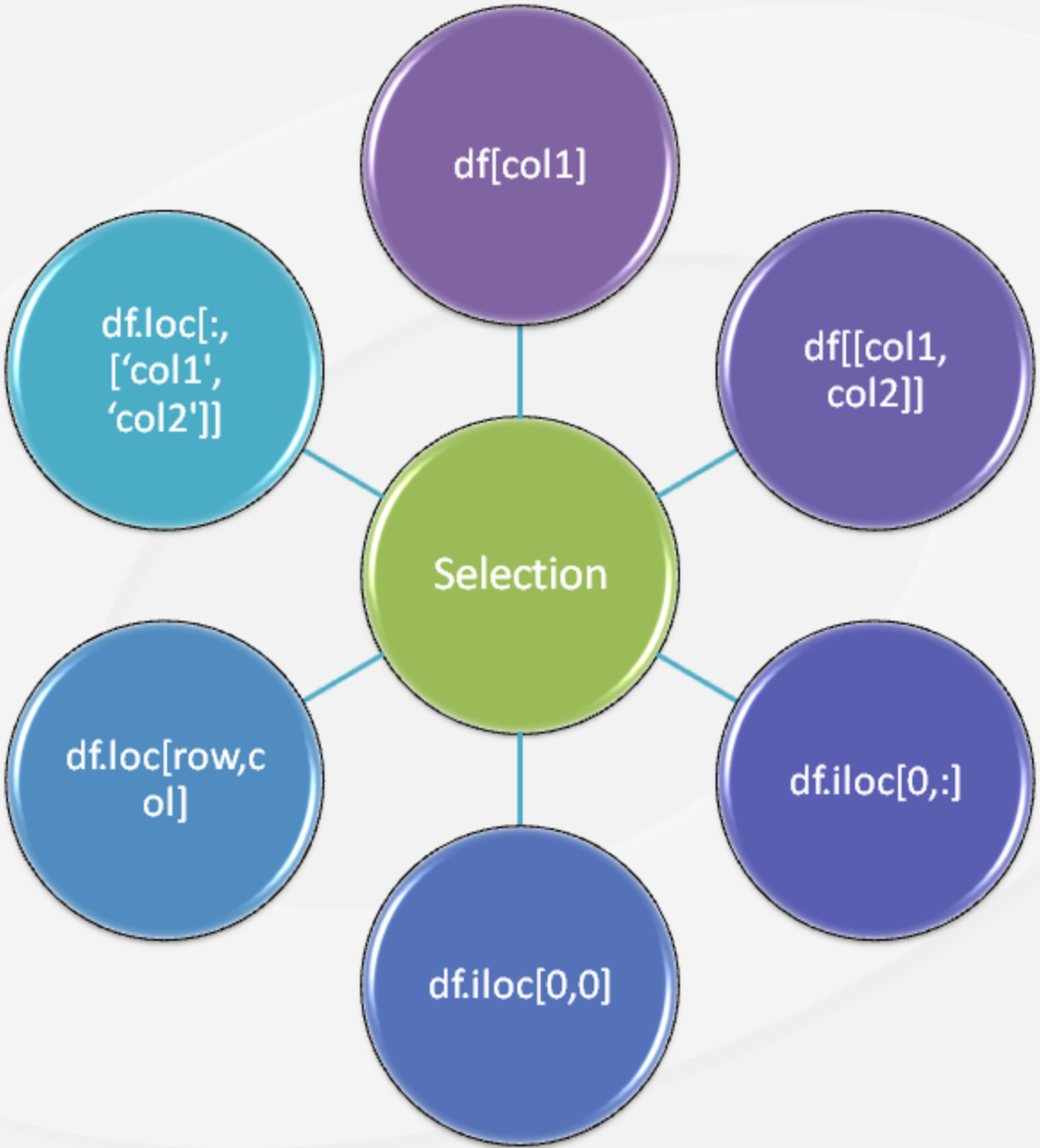
Selection



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Filter, Sort, and Groupby

OBJECT™ TECHNOLOGIES

Filter

- df[df[col] > 0.5]
- df[(df[col] > 0.5) & (df[col] < 0.7)]
- df.apply(functionname)

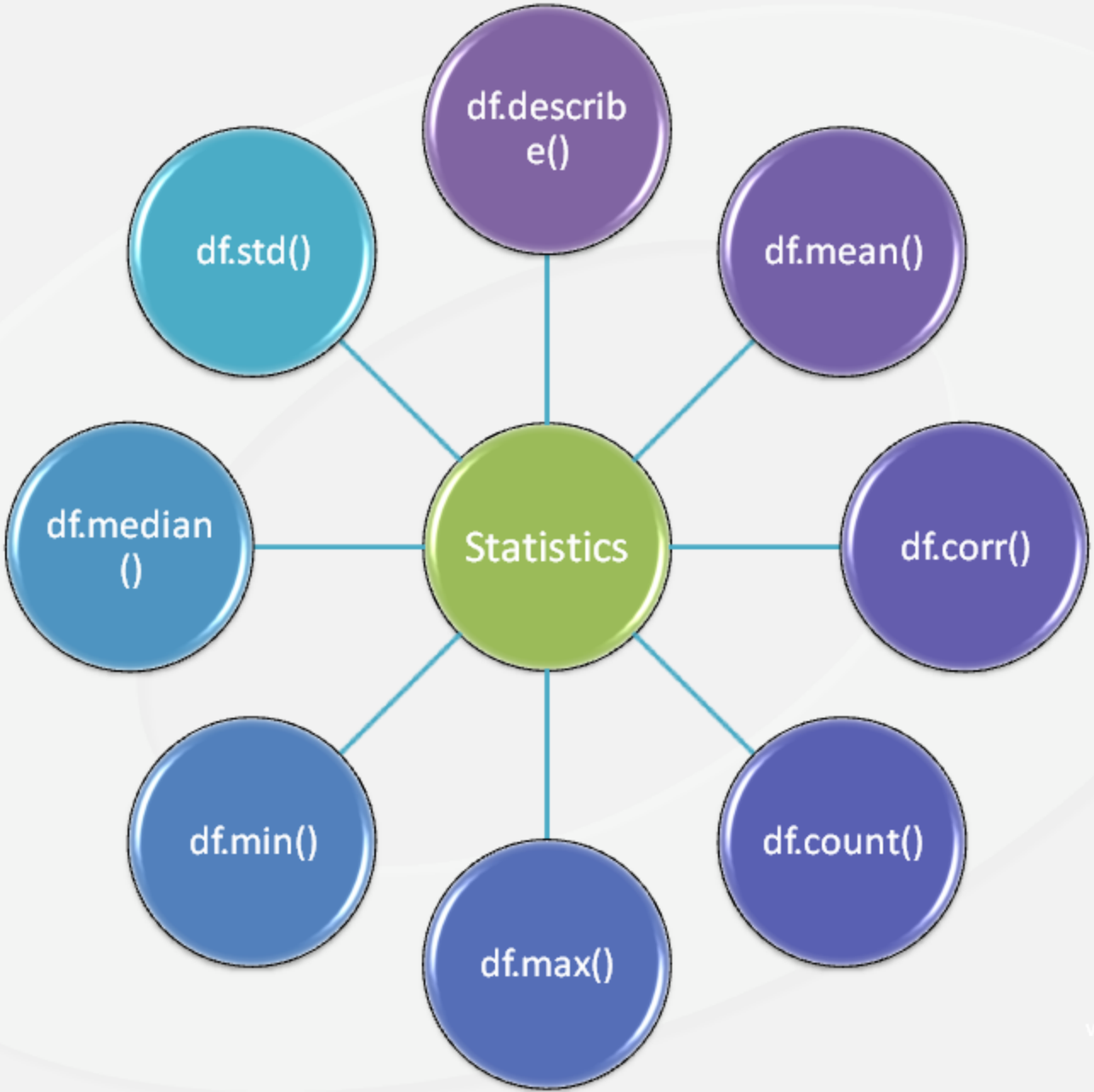
Sort

- df.sort_values(col1)
- df.sort_values(col1,ascending=False)

Group by

- df.groupby(col1)
- df.groupby([col1,col2])

Statistics



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Join/Combine

OBJECT™ TECHNOLOGIES

```
df1.append(df2)
```

```
pd.concat([df1, df2], axis=1)
```

```
pd.merge(df1,df2,on=col1,how='inner')
```

Data Cleaning

OBJECT™ TECHNOLOGIES

General

- df.columns = ['a','b','c']
- df.set_index('column_one')
- s.astype(float)

null values

- pd.isnull()
- pd.notnull()
- df.dropna()
- df.dropna(axis= 1)
- df.dropna(axis=1,thresh=n)
- df.fillna(x)
- s.fillna(s.mean())

replace

- s.replace(1, 'one')
- s.replace([1,3],['one', 'three'])

Rename

- df.rename(columns=lambda x: x+1)
- df.rename(columns={'old_name': 'new_ name'})

Summary

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Thank You